# B1 plus full-document sparse hybrid: Methodology and Evidence

CS781 Capstone – Krishna K Karthikeya G N

12 November 2026

**Status.** Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

## Research question

The neural baseline struggled with long documents, so add a full-text branch that can retain sparse statute phrases wherever they appear. The weight was chosen on calibration data, then held fixed for dev and test. This is an ensemble of an adapted Phase 1 model and a sparse classifier, not a standalone neural adaptation.

## Architecture and mechanism

The official-train sparse classifier reads the full document while the B1 LoRA branch supplies contextual probabilities. A fixed probability mixture uses 0.9 sparse and 0.1 B1 weight; thresholds are fitted on grouped official-train calibration.

## Measured result

| | |
|---|---|
| Scored submission | 1016346, 1016350 |
| Codabench macro-F1 | 0.3949 |
| Archived local metric | 0.3909 official-dev macro-F1 |

Official-dev macro-F1 0.3909 and Codabench 0.3949 both materially exceed the LoRA branch. The record supports the value of full-document lexical coverage, while the mechanism remains an inference from the comparison.

## Limitations and next check

Course-method eligibility of a neural-plus-sparse hybrid was not confirmed in the archived report; keep that distinction visible. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

## Source and reproducibility

Original source modules: phase5_hybrid_blend.py, phase5_prepare_hybrid_probs.py, phase5_b1_train.py, run_logistic.py, run_sparse.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

## Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.